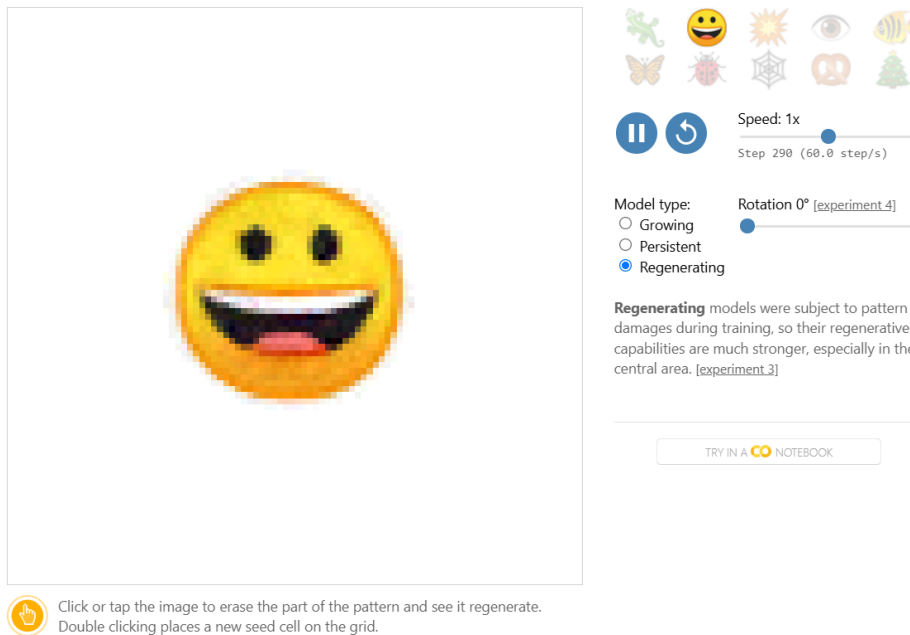


NEURAL CELLULAR AUTOMATA (NCA)

First a brief into what NCA actually is:

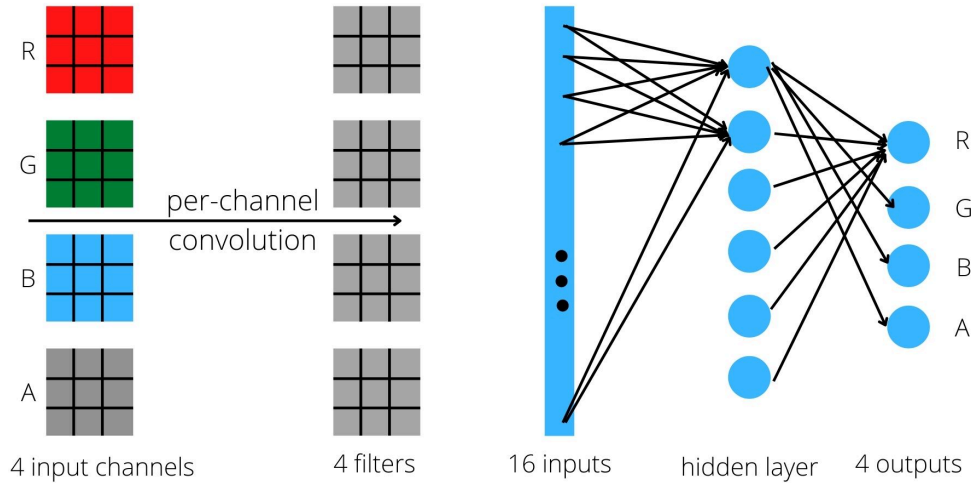
Growing Neural Cellular Automata

Differentiable Model of Morphogenesis



The fundamental basis of this big smiley is the same as Game of Life... each pixel is only aware of the state of its neighbours... What's special about this is that, if at all we erase certain pixels, the image gets repaired... and the original shape and the new shape generated usually have a high correlation... Given the percentage of pixels deleted is not a lot...

A small neural network is set into place for this to happen.. Which we are gonna discuss



The 4 input channels represent the main cell, as well as its neighbours... This diagram considers only 4 channels because of its simplistic nature, but many hidden channels can be added in order to improve functionality and accuracy as well...

RGB represents the color of the cell basically and alpha is a special channel used in NCA... which represents the state of the cell :

The Living Threshold: The NCA uses the Alpha channel to determine if a cell is alive. If a cell's Alpha value drops below a certain mathematical threshold (usually 0.1), the model considers that cell dead, empty, or uninitialized.

Growth and Regeneration: When the NCA is trying to grow a shape or heal a damaged section of the grid, it actively manipulates this Alpha value to "turn on" empty neighboring cells, transforming them from transparent void into solid, colored parts of the organism.

These channels then pass through different filters (4 here) and hence 16 inputs are generated in this example...

Each filter is used to extract a different type of information... A filter is basically a 3x3 grid , which convolutes with the channels in order to produce new inputs... which go through a hidden later in order to produce outputs

The output of the network is a 4-dimensional vector $(\Delta R, \Delta G, \Delta B, \Delta A)$.

Crucially, the network does not output the *new* state of the cell; it outputs a **proposed update** (a delta). The model uses a residual connection to update the cell for the next time step, integrating the changes over time:

$$S_{t+1} = S_t + \Delta S$$